

flashjet

GPU-native jet clustering with the full merge history
designed for reclustering inside training loops, on padded tensors that stay on the GPU

Chirayu Gupta^{1,*} Sitian Qian² Alexandre De Moor¹

¹Vrije Universiteit Brussel (BE)

²Peking University (CN)

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* speaker

- 1 Jet clustering, and flashjet
- 2 Correctness
- 3 Speed — the jet regime
- 4 Downstream: the clustering tree as tagger input

Jet clustering, and flashjet

Reconstruction is moving to accelerators — clustering has not

1. Where the field is

- GPUs in the CMS High Level Trigger since Run 3
- **~30 %** of HLT reco offloaded → **~25 %** less HLT time
- pixel local reco · tracks · vertices · calo local reco

2. What is missing

Jet clustering is **still on the CPU**.

3. Why it stayed there

- sequential recombination is **IRC safe** — that is why it won
- but each merge **depends on the previous one**
- serial dependence is what a GPU does worst

4. Why that costs us now

- **reclustering inside a training loop**
- scanning the **jet radius R**
- **substructure** on demand

The tensors are already on the GPU — each of these sends them back to the host and returns, once per call.

What it does

- generalised- k_t — anti- k_t , k_t , C/A — **transcribed from FastJet and validated against it**
- **many jets or events per launch**
- returns jets, the **full merge history**, and substructure **as tensors**

⇒ built for **reclustering inside training loops**, on padded $(B, N, 4)$ tensors that **stay on the GPU**

Why it can be done at all

- Cacciari–Salam lemma: the smallest d_{ij} in the whole event is always some particle's **geometric** nearest neighbour
 - so each particle tracks only **its own** NN, and one merge invalidates $O(1)$ of them
- ⇒ $O(N^2)$, **not** $O(N^3)$ — the serial chain stays, but each step is cheap and parallel

In the ML ecosystem

- a **PyTorch** call — tensors in, tensors out
- no host $\leftrightarrow$ device copy in a training loop

Two regimes: *event-based* ($N \sim$ hundreds) and *jet-based* ($N \sim$ tens, B large). **This talk is the second.**

Written in **Triton**: autotunes for whatever GPU it finds — no separate CUDA build.

One exponent, three algorithms

The generalised- k_t distance measure:

$$d_{ij} = \min(p_{Ti}^{2p}, p_{Tj}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \quad \Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$$
$$d_{iB} = p_{Ti}^{2p}$$

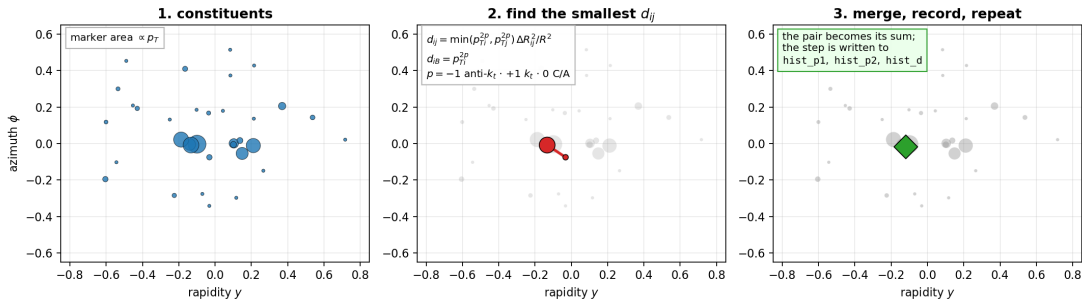
Merge the pair with the smallest d_{ij} ; if the smallest distance is a d_{iB} , call i a jet and remove it.

Only the exponent p changes:

p	algorithm	merges first	tree ordered by	used for
-1	anti- k_t	the hardest pairs	—	finding jets
0	Cambridge–Aachen	the closest pair	angle	grooming, Lund plane
+1	k_t	the softest pair	relative p_T	exclusive subjets

anti- k_t answers *which particles form a jet*. The other two answer *how that jet came apart* — and the **ordering is what the substructure reads**.

Sequential recombination — and what flashjet keeps from it



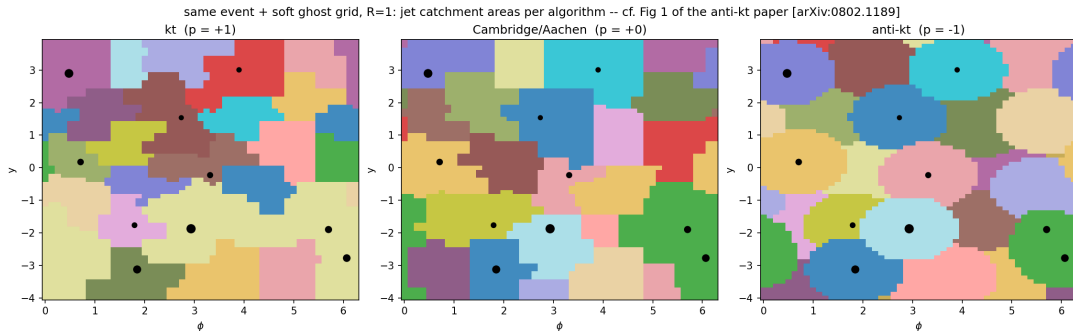
Two distances, computed for every pair:

- d_{ij} — *pair* distance, particle i to particle j
- d_{iB} — *beam* distance, particle i to the beam

Then repeat until nothing is left:

- ① find the smallest of all distances
- ② if it is a d_{ij} : **merge** i and j
- ③ if it is a d_{iB} : call i a **jet**, remove it

Both distances were defined on the previous slide; the exponent p there selects which algorithm this loop runs.



- jet catchment areas, all three algorithms
- clustered by **flashjet** itself

- k_t , C/A: **ragged**, fluctuating boundaries
 - anti- k_t : **circles of area πR^2** on each hard particle
- $\Rightarrow$ the algorithm's defining result, **reproduced without FastJet**

It returns the tree, not just the jets

Clustering a batch returns **two** things — the jets, and the record of how they were built.

field	shape	what it holds
jet_idx	(B, N)	which jet each particle ended up in
n_jets	$(B,)$	how many jets were found
hist_p1, hist_p2	(B, N)	which pair merged , at each step
hist_child	(B, N)	what they merged <i>into</i>
hist_d	(B, N)	the distance d_{ij} at that step

The tree is the product, not a by-product

The bottom three rows are the **complete binary merge tree**. Groomed mass, z_g , R_g , Lund coordinates and exclusive subjets are not separate algorithms — they are **coordinates on that one tree**. Compute it once, read them all off, **no second clustering pass**.

Three things are read off that tree — F1, F2, F3 — and then we look at it on a real jet.

What is implemented

Clustering

- generalised- k_t : anti- k_t , k_t , C/A
- merge history recorded in-kernel
- many jets clustered in one batched call
- scales from single jets to whole events

Substructure — reads of the history

- **F1** exclusive k_t subjets
- **F2** soft drop / mMDT / mass-drop
- **F3** primary Lund coordinates

All three are pure-torch post-reads: no kernel changes, CPU/GPU identical, negligible cost.

What it does. Undo the last merges of the sequence to expose exactly n subjects — or stop at a d_{cut} .

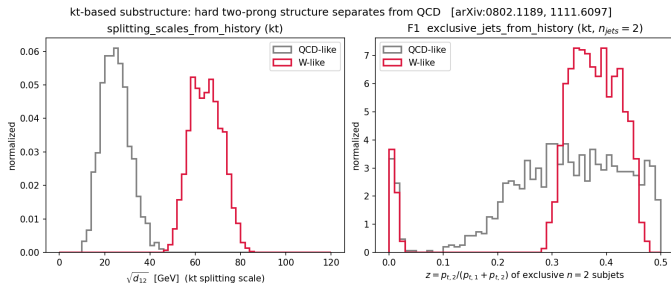
How. The history is already a binary tree. Walk back up it $n - 1$ steps; no reclustering.

Why k_t . k_t merges the softest pair first, so the *last* merges are the hard prongs — exactly what a 2-prong decay leaves behind.

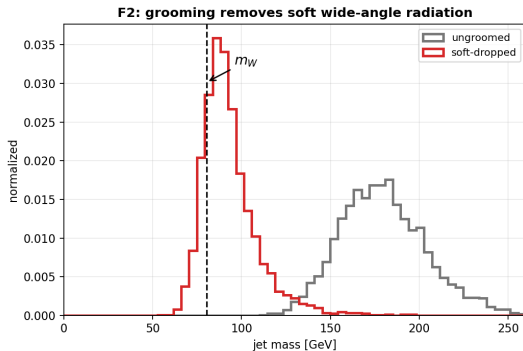
The splitting scales come straight from `hist_d`:

$$\sqrt{d_{12}}, \sqrt{d_{23}}, \dots$$

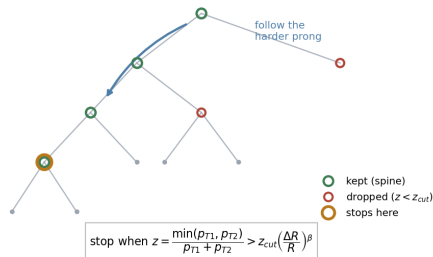
$$\text{with } d_{ij} = \min(p_{Ti}^2, p_{Tj}^2) \frac{\Delta R_{ij}^2}{R^2}$$



Toy W-like (2-prong) vs QCD-like fat jets. **Left:** the k_t splitting scale $\sqrt{d_{12}}$ separates cleanly. **Right:** the momentum balance z of the two exclusive subjects — the W sits near $z \sim 0.4$, QCD spreads to small z .



the soft-drop walk on the C/A tree



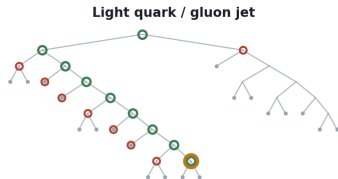
Stop when
$$\frac{\min(p_{T1}, p_{T2})}{p_{T1} + p_{T2}} > z_{\text{cut}} \left(\frac{\Delta R_{12}}{R} \right)^\beta$$

Walk down the C/A tree along the harder branch, discarding the softer prong until that condition is met. $\beta = 0$ recovers mMDT.

The walk is $O(\log_2 n)$ on the stored tree — it never touches the constituents again, so grooming is essentially free once clustered.

C/A merge trees of real jets — root at top, constituents at the bottom

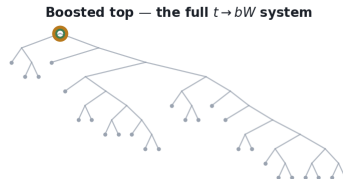
ATLAS Top Tagging Open Data · vertical = declustering depth · horizontal = angular ordering



a long spine: soft prong after soft prong
is stripped, and the mass collapses
 $n_{\text{drop}} = 8$ $m: 80 \rightarrow 1 \text{ GeV}$



the very first split is already balanced,
so nothing is groomed away
 $n_{\text{drop}} = 0$ $m: 80 \rightarrow 80 \text{ GeV}$



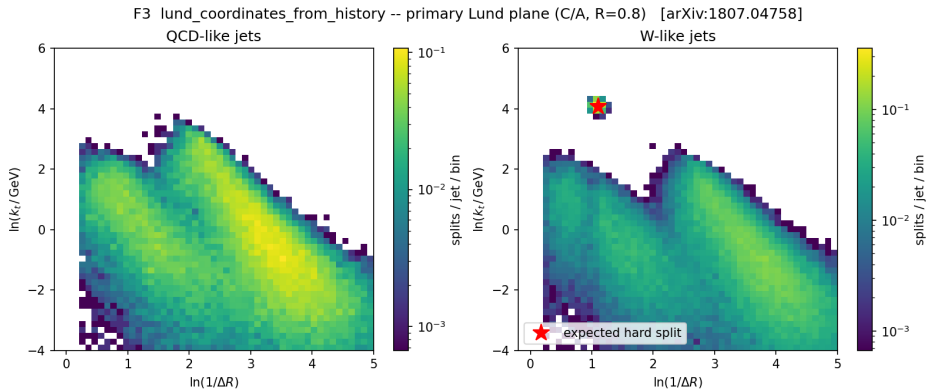
also stops at once, but on a wider split,
so the whole decay is kept
 $n_{\text{drop}} = 0$ $m: 170 \rightarrow 170 \text{ GeV}$

● soft-drop spine ● dropped prong ● grooming stops here ● constituent

A jet without a decay has nothing to stop on. The light-quark/gluon jet is a long *staircase*: soft prong after soft prong is stripped, and its mass collapses to nothing.

ATLAS Open Data, C/A with soft drop ($z_{\text{cut}}=0.1$, $\beta=0$). **The number of drops is itself a discriminant** — and it is read straight off the stored tree, at no extra cost.

A real decay stops at once. Both top jets halt on their very first split, because it is already hard and balanced — so the mass survives.



What. Every primary splitting emits $z = \frac{p_{T2}}{p_{T1} + p_{T2}}$, ΔR , $k_t = p_{T2} \Delta R$, and the plane coordinates $(\ln \frac{1}{\Delta R}, \ln k_t)$.

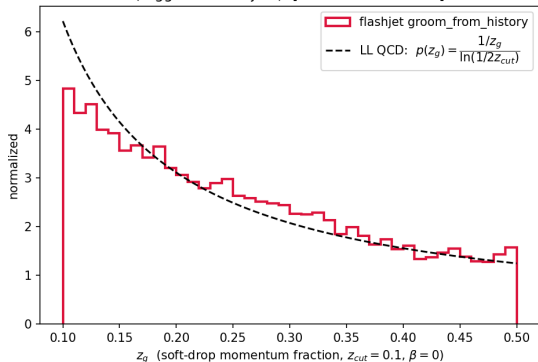
QCD fills the soft-collinear region smoothly; the W-like sample adds a **localised hard splitting** exactly where the decay is predicted to sit (star).

How. One pass down the primary branch — each node is one point in the plane.

Check. The d -channel ties *exactly* to the splitting scales computed independently.

Correctness

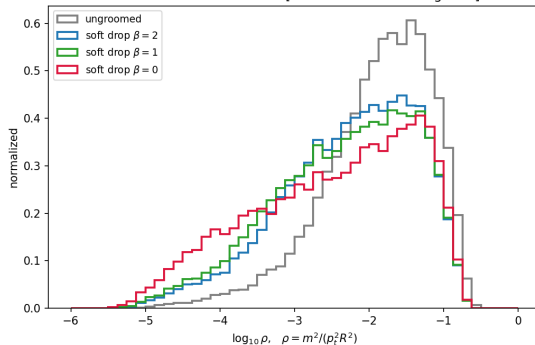
groomed splitting fraction vs the analytic prediction
(tagged 72% of jets) [cf. arXiv:1402.2657]



Soft-drop z_g follows the $1/z$ form predicted at leading-log, recovered from the C/A history.

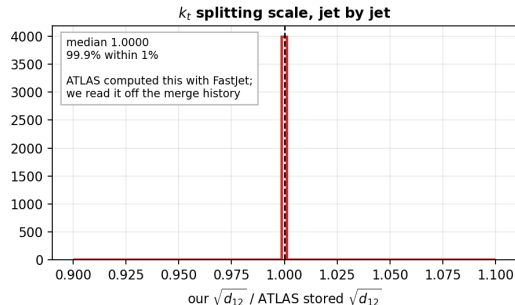
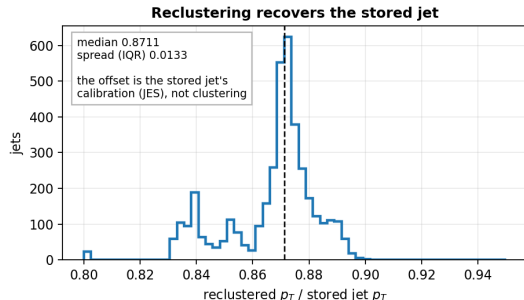
Toys first: on generated input the answer is known analytically, so anything off the curve is a real bug. Real-data checks follow. Toy-shower input tests the decoders, not an event generator.

groomed jet mass: stronger grooming (smaller β) pushes the soft-radiation mass down [cf. arXiv:1402.2657 Figs 3-4]



Groomed-mass ordering in β : larger β grooms less, so the mass peak moves up.

Jet-by-jet closure against ATLAS-stored quantities



Left. Reclustering the stored constituents recovers the stored jet with a **0.013 spread**. The 0.87 offset is the stored jet's **calibration**, applied after clustering — it is present before we touch anything.

Right. The k_t splitting scale $\sqrt{d_{12}}$ against the value **ATLAS computed with FastJet** and stored: median **1.0000**, **99.9 %** within 1%. We read ours straight off the merge history.

vs FastJet — a 150-point grid

3 algorithms $\times$ 5 radii $\times$ 5 multiplicity bins $\times$ 2 samples, 20 000 jets each.

100.000 % jet-count agreement at **every** point.

Leading-jet p_T within 10^{-4} at 148/150; median difference $\sim 7 \times 10^{-8}$ — float32 precision.

vs ATLAS's own reconstructed jets

Cluster the ~ 600 stored calorimeter clusters ourselves; compare to the jets ATLAS reconstructed and published.

100.0 % n_{jets} match, **10.96 vs 10.96** jets/event.

The event regime, where the problem is $O(N^2)$.

Two *independent* references: FastJet for the jets, a separate **NumPy tree-walk** for every substructure decoder. The second test closes the algorithm against **the experiment's published output** — not against another implementation of the same algorithm.

Per-point tables and the R /multiplicity scans are in backup.

Speed — the jet regime

Data — ATLAS Open Data, freely presentable

sample	jets	$\langle n_{\text{const}} \rangle$
boosted top	30 000	59.2
QCD (light q/g)	30 000	52.4

Large- R jets with calorimeter-cluster constituents. The grid is **3 algorithms** $\times$ **5 radii** $\times$ **5 multiplicity bins**, 20 000 jets per bin.

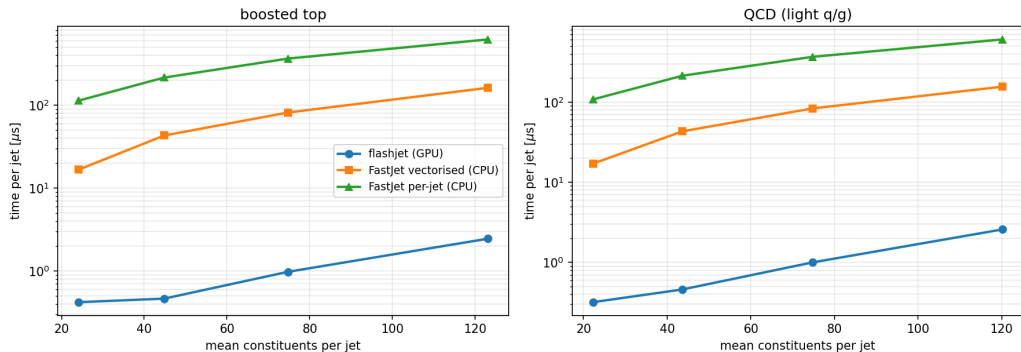
Method

- 1 extract constituents once, save;
- 2 cluster on GPU, time it, save the *same* events;
- 3 cluster those saved events with FastJet;
- 4 **check they agree** — else the timing is meaningless.

Both clusterers read the **same saved file**. Speedups are quoted against the **vectorised** FastJet interface — the faster, fairer baseline.

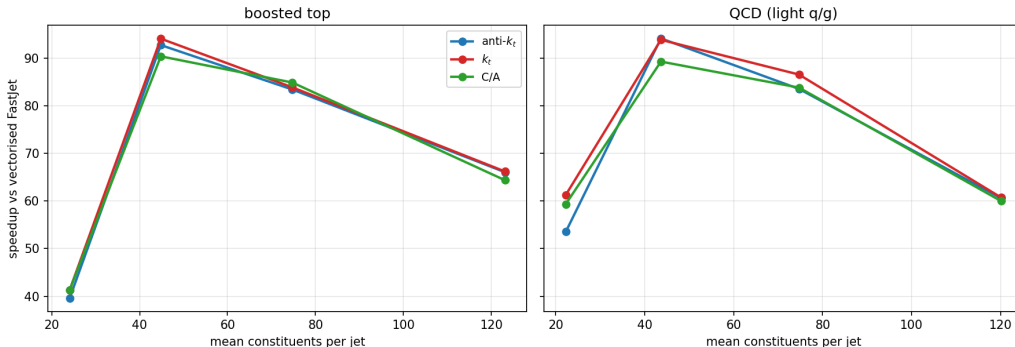
One timed unit = **one large- R jet** (24–123 constituents), batched. Event-regime timing is not in this talk.

ATLAS Top Tagging Open Data — anti- k_t $R = 1.0$



Log scale — each gridline is $10\times$. flashjet stays nearly flat while both FastJet interfaces climb steeply with multiplicity: **$0.35\text{--}2.7\ \mu\text{s}$ per jet** against **$17\text{--}155\ \mu\text{s}$** for vectorised FastJet, and **$110\text{--}620\ \mu\text{s}$** per jet one at a time. Benchmarked on an NVIDIA V100-class GPU. **Per jet, batched** — the GPU clusters the whole batch at once, so this is amortised cost, not single-jet latency.

Speedup vs multiplicity — $R = 1.0$ (all points 100% n_{jets} agreement)



All three algorithms, both samples, $R = 1.0$. **39–99 $\times$** over vectorised FastJet across the grid (median **64 $\times$**), and every point has **100 %** n_{jets} agreement. Busier jets $\rightarrow$ bigger win — the regime that matters for boosted-object physics.

Using it

```
import flashjet

out = flashjet.cluster(p4, mask, R=1.0, algorithm="antikt")
```

argument	type	
p4	$(B, N, 4)$ tensor	(p_x, p_y, p_z, E) ; CPU or CUDA
mask	(B, N) bool	marks real constituents in a padded batch
R	float	jet radius
algorithm	str	"antikt" "kt" "cambridge"
p	float	generalised- k_t exponent (overrides algorithm)

out.n_jets	# (B,)	jets found
out.jets_p4(p4)	# (B,J,4)	jet four-momenta
out.splitting_scales()	#	$\sqrt{d_{12}}$, $\sqrt{d_{23}}$, ...
out.exclusive_jets(n_jets=3)	# F1	exclusive k_t subjects
out.groomed_jets(p4, R, z_cut=0.1, beta=0.0)	# F2	soft drop / mMDT
out.lund_coordinates(p4, R)	# F3	primary Lund plane

The same call runs unchanged on CPU and GPU — the backend is chosen from the problem size, not by the caller.

Downstream: the clustering tree as tagger input

Does the tree help a tagger? — setup

The question. Gouskos & Maier [PLuM, 2605.26821] report that explicit Lund-plane splittings help a transformer tagger. We test that on a **full 10-class** training.

Model. Particle Transformer [2202.03772] — 8 encoder + 2 class-attention blocks, 8 heads, embed 128; **2.144 M** parameters.

Data. JetClass, 10 classes, 100 M training jets, **~20.05 M** test jets, 128 particles/jet.

Training. Ranger, lr 10^{-3} , batch 512, 10^6 **iterations** to convergence. All arms read **identical shards** in the same order; features are computed **live** by flashjet inside the loop.

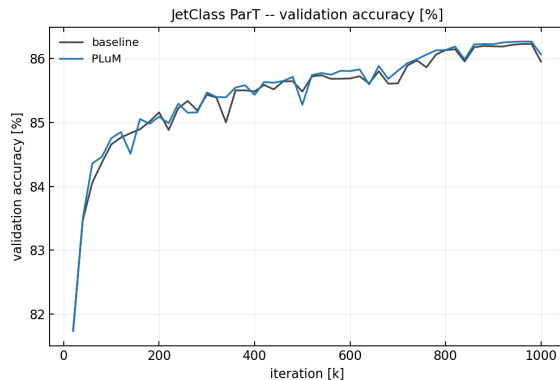
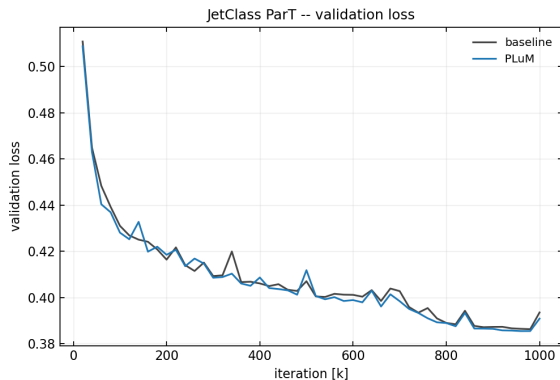
Our baseline reproduces the paper

	ours	published
test accuracy	86.21 %	~86.1 %
parameters	2,193,930	2.19 M

Like-for-like — so a difference between arms is the *feature*, not a broken baseline.

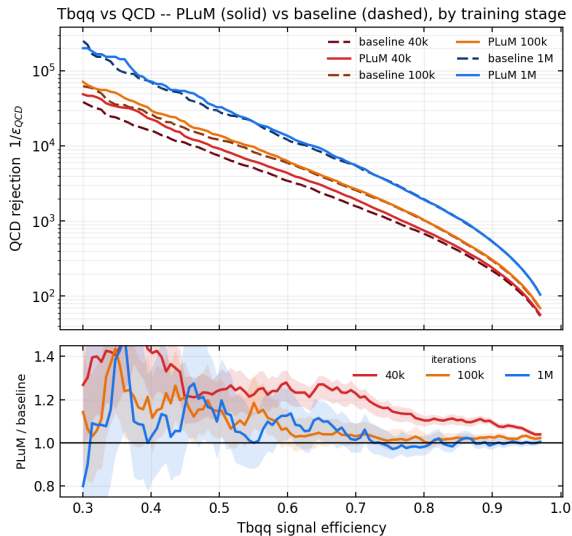
Shown here: PLuM — 48 k_t -ordered Lund splittings [$\ln k_t, \ln 1/\Delta R, \ln z$], concatenated as extra tokens (176 vs 128). Two other encodings were tried — **subject** (n -body mass, p_T fraction, n_{const}) and **CA5** (per-particle C/A branch points); both null-or-worse (backup). Bootstrap errors, 16–84th pct. **One seed per arm.**

Both arms train identically



Validation loss and accuracy over the full 10^6 iterations. The curves are **on top of each other** — the feature block does not destabilise training, and it does not accelerate it either. Both converge to $\sim 86.2\%$.

Top tagging: the gain closes with training



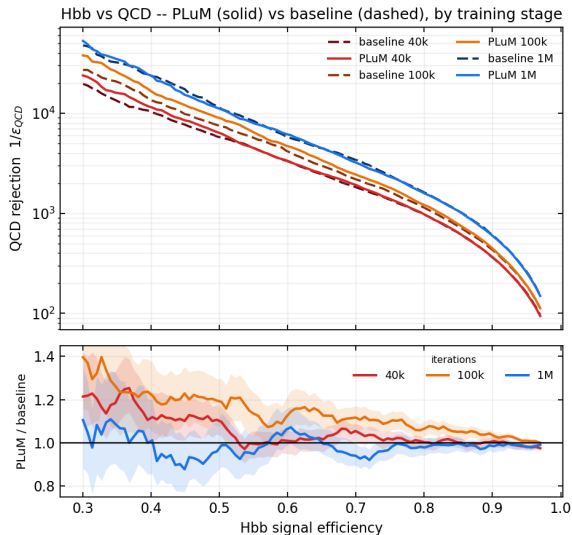
$t \rightarrow bqq$ vs QCD.

Top panel: full ROC at three training stages.
Bottom: **PLuM / baseline** rejection ratio.

At **40k** the ratio sits **10–25 % above 1**. By **1M** it has collapsed onto the line.

Bands: unpaired bootstrap of the test set, 16–84th percentile. The low-efficiency end is noisy because rejection there is $\sim 10^5$.

$H \rightarrow b\bar{b}$: the paper's best channel

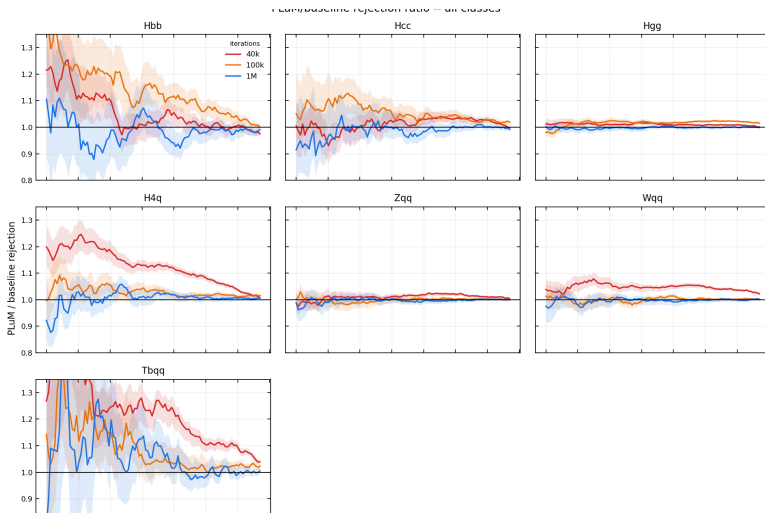


$H \rightarrow b\bar{b}$ vs QCD — where PLuM report their largest gain.

We **reproduce it early**: +20 % at 100k. At **1M** the ratio is **at or below 1**.

Their result is obtained in *binary* mode on a fast simulation **without secondary-vertex inputs**; ours is a 10-class training. The early-training agreement is real — the disagreement is about what survives to convergence.

All seven signal classes — the same story



PLuM/baseline rejection ratio; unpaired-bootstrap bands.

Test accuracy ~ 20.05 M jets

	40k	100k	1M
baseline	83.431	84.632	86.212
PLuM	83.428	84.697	86.206
Δ	-0.003	+0.065	-0.006

What we see

An early gain in every class, gone by 1 M.
Even Wqq — the *fewest* splittings — gains at $\sim 9\sigma$ early: **not confined to complex or *b*-jets**. At convergence: **null-or-worse**.

What we do *not* claim

- **No mechanism** — *b*-specificity excluded.
- **One seed per arm** — a second seed's spread exceeded the effect.
- Their Hbb reproduced at 100k, **not** at convergence.

- 1 **Clusters on the GPU and keeps the full merge history**, so substructure — exclusive subjects, soft drop, Lund coordinates — comes free as array reads. Built for **reclustering inside a training loop**, on tensors that never leave the GPU.
- 2 **Gives the same jets as FastJet**. 150/150 grid points at 100 % n_{jets} agreement, across anti- k_t/k_t /C-A, $R = 0.4\text{--}1.2$ and 24–123 constituents per jet.
- 3 **Reproduces the experiment's own reconstructed jets** exactly, at ~ 600 clusters per event.
- 4 **39–99 $\times$ faster** than vectorised FastJet **in the jet regime** (one large- R jet per timed unit, batched), and the advantage *grows* with jet complexity. Cost is independent of R and of algorithm.
- 5 Still a **prototype**, with meaningful headroom left.

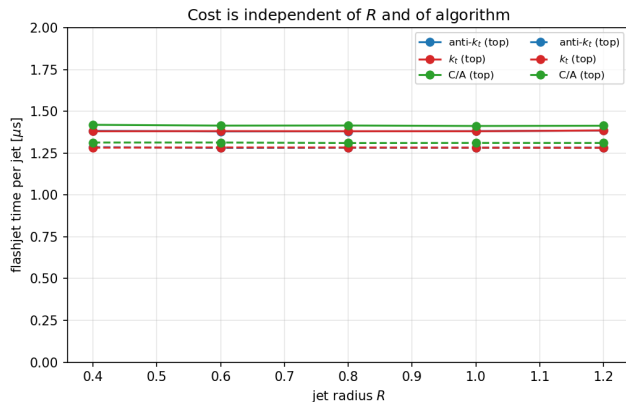
Thank you

Questions?

Benchmarks: ATLAS Open Data (CC0).

Physics closures: analytic predictions on toy showers.

Backup

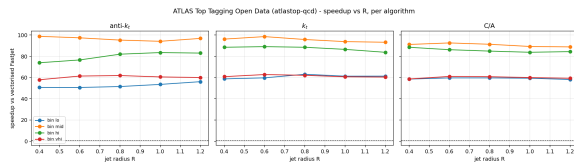
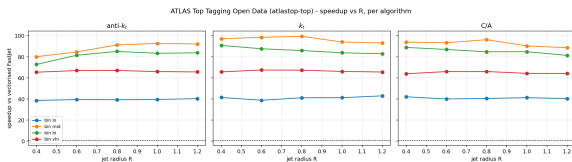


Across $R = 0.4 \rightarrow 1.2$: a **0.3 %** spread over a $3\times$ range in R .

R changes *which* pairs merge, not *how many* distances are computed.

Across algorithms: anti- k_t , k_t and C/A agree to within $\sim 4\%$ at every multiplicity.

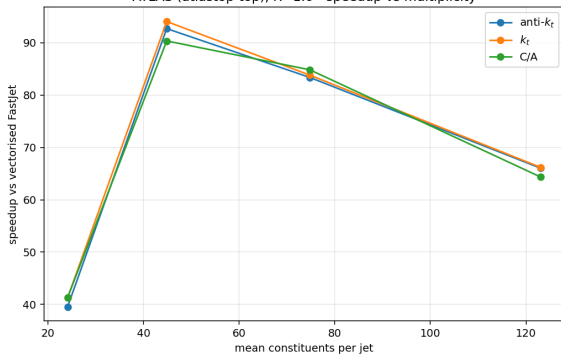
k_t and C/A cost the same as anti- k_t .



boosted top
Flat in R for all three algorithms; ordering is by multiplicity bin.

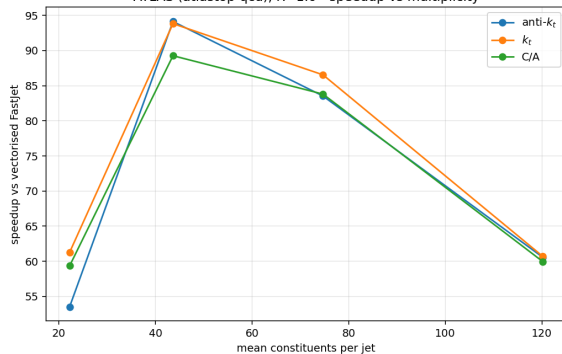
QCD (light q/g)

ATLAS (atlastop-top), R=1.0 - speedup vs multiplicity



boosted top

ATLAS (atlastop-qcd), R=1.0 - speedup vs multiplicity



QCD (light q/g)

Backup — the numbers (jet regime)

sample	alg	$\langle n \rangle$	flashjet [$\mu\text{s}/\text{jet}$]	FastJet vec. [$\mu\text{s}/\text{jet}$]	speedup
top	anti- k_t	24.1	0.427	16.5	39×
top	anti- k_t	44.8	0.464	43.0	93×
top	C/A	74.7	1.000	84.8	85×
top	C/A	123.1	2.565	165.1	64×
qcd	anti- k_t	22.2	0.346	17.5	51×
qcd	anti- k_t	43.6	0.463	45.7	99×
qcd	anti- k_t	74.7	1.215	89.8	74×
qcd	anti- k_t	120.2	2.673	154.9	58×

39–99× over vectorised FastJet (median 64×), with 100% jet-count agreement everywhere. **One timed unit = one jet**, batched. Benchmarked on an **NVIDIA V100-class GPU**.

$R = 1.0$ rows shown. Against the per-jet FastJet interface the gap is roughly *two orders of magnitude*.

Backup — where the time goes

Nsight Systems — share of GPU time

kernel	% GPU	avg/call
_cluster_large_kernel	97.6	5.47 ms
_decode_kernel	1.5	82 μ s
torch glue	< 1	—

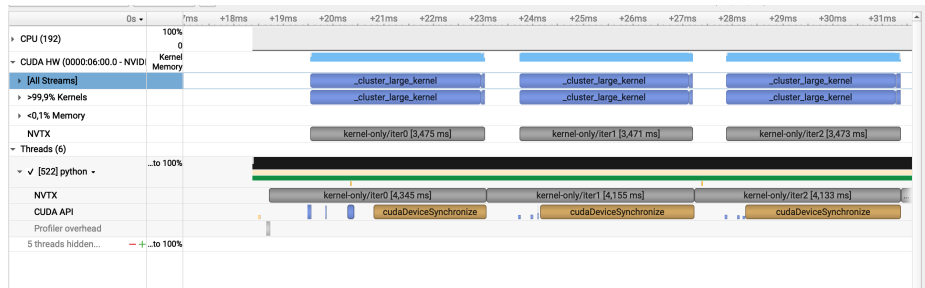
No hidden overhead — essentially all the time is real clustering work, and the memory row of the timeline is empty.

Nsight Compute — hardware counters

metric	value
DRAM throughput	0.02 %
Compute (SM) throughput	15.4 %
Registers / thread	168
Theoretical occupancy	18.75 %
Achieved occupancy	6.25 %
Waves per SM	0.32

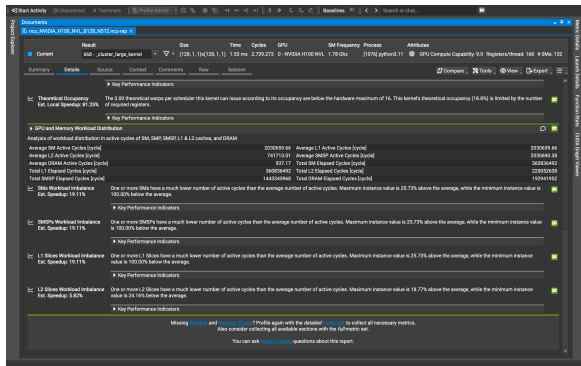
Not memory- or compute-bound — the GPU is simply **not filled**.

Backup — the GPU timeline



Nsight Systems, three consecutive iterations. Each block is one clustering call; the tool labels the rows “>99.9 % Kernels” and “<0.1 % Memory”. The kernels run back to back with nothing in between and the memory row is empty — no host/device shuffling while clustering runs. That is what makes the speedup real rather than an artefact of where the timer was started.

Backup — the profiler names the bottleneck



The tool's own speedup estimates:

finding est. speedup

Theoretical occupancy (registers) 81 %

Achieved occupancy 67 %

SM workload imbalance 19 %

"This kernel grid is too small to fill the available resources on this device, resulting in only 0.32 full waves across all SMs."

Grid size **128 blocks** on a **132-SM** device — fewer blocks than the GPU has processors.

~**1.7–1.8×** more available from launch configuration and register budget — **not** from the algorithm.

Backup — occupancy detail

► Occupancy

Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution indicate typically highly imbalanced workloads.

Theoretical Occupancy [%]	18.75	Block Limit Registers [Block]	3
Theoretical Active Warps per SM [warp]	12	Block Limit Shared Mem [Block]	6
Achieved Occupancy [%]	6.52	Block Limit Warp [Block]	16
Achieved Active Warps Per SM [warp]	4.00	Block Limit SM [Block]	32
Cluster Occupancy [%]	0	Block Limit Barriers [Block]	32
Max Active Clusters (Cluster)	0	Max Cluster Size [Block]	8
Overall GPU Occupancy	0		

ⓘ The difference between calculated theoretical (18.75%) and measured achieved occupancy (6.52%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See [§ 4.2.4.2.2.1. Occupancy](#) for more details on optimizing occupancy.

► Key Performance Indicators

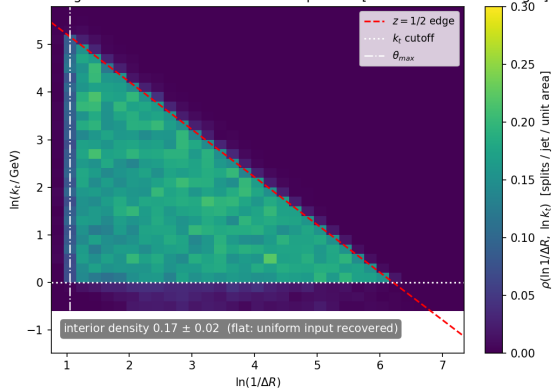
ⓘ The 5.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 16. This kernel's theoretical occupancy (18.8%) is limited by the number of required registers.

► Key Performance Indicators

[illegible]

Register pressure (168/thread) caps theoretical occupancy at 18.75 %; achieved is 6.25 %. With 0.32 waves/SM the kernel does not fill the GPU even once, while DRAM throughput is 0.02 % — this is parallelism starvation, not a bandwidth problem. Both are configuration inherited from smaller GPUs, not properties of the algorithm.

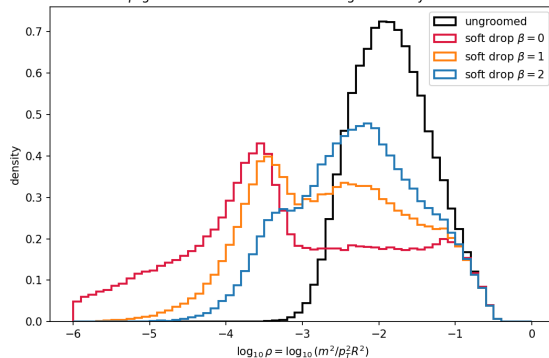
primary Lund plane of the toy shower (C/A, R=0.4):
m-in-triangle emissions come back as a flat plateau [cf. arXiv:1807.04758 Fig 2]



The Lund plane is populated **uniformly** inside the kinematic **triangle** — the leading-log expectation.

Independent NumPy tree-walks pin every decoder, and the d -channel of the Lund output ties *exactly* to the splitting scales computed separately.

Soft-drop β -ordering on REAL QCD jets (164292 jets, $z_{cut} = 0.1$)
smaller β grooms harder — same ordering as the toy-shower closure



The soft-drop β -family orders as predicted: larger β grooms less.

Backup — Input A: the toy fat jets

Purpose. QCD vs W is a *known* truth, so any observable that fails to separate them — or lands off the analytic curve — is a real bug.

Two single-fat-jet samples at $p_T \approx 500$ GeV, $R = 0.8$:

- **QCD-like:** one hard collinear core (92 % of p_T , $\sigma_{y,\phi} = 0.06$) + soft wide-angle radiation (8 %, $\sigma = 0.30$)
- **W-like:** two hard prongs with pair mass $m = \sqrt{z(1-z)} p_T$, $\Delta R = 80.4$ GeV, $z \in [0.30, 0.45]$, random orientation, + 6 % soft contamination

Each “spray” is n massive pions with exponential p_T fractions:

```
frac = rng.exponential(1., n); frac /= frac.sum()
pt    = pt_total * frac
y, phi = rng.normal(y0, sigma, n), rng.normal(phi0, sigma, n)
mt     = np.sqrt(M_PI**2 + pt**2)
```

1500 events per sample. `flashjet` has **no event generator** — it clusters four-vectors; these toys live in the plot scripts, outside the library.

Backup — Input B: the toy parton shower

A **primary, fixed-coupling, leading-log** shower: emissions sampled **uniformly in the Lund triangle** with density $\bar{\alpha}$ per unit $(\ln 1/\theta, \ln k_t)$ area, off a hard spine.

$$\theta = e^{-u}, \quad k_t = e^v, \quad z = \frac{k_t}{p_{T0} \theta}$$

keep $v \leq \ln(p_{T0}/2) - u$ (i.e. $z \leq \frac{1}{2}$), $k_t > k_t^{\min}$

Why this is the right test

The generator's density is *uniform in the triangle by construction*, so the leading-log predictions — the $1/z$ form of z_g , the β -ordering of the groomed mass — are unambiguous. Landing on them tests the **decoders**, not the shower.

Jet areas use the anti- k_t -paper construction: one hard event plus a dense grid of **infinitely soft ghosts** ($p_T = 10^{-8}$); the ghosts each algorithm sweeps up trace its catchment area.

$\bar{\alpha} = 0.25$, $p_{T0} = 1$ TeV, $R = 0.4$, 20 000 showers; fixed seed, fully re-runnable.

Backup — cost of each feature arm

Wall clock per training step, relative to the baseline:

arm	cost
PLuM	1.59×
subjet	1.42×
CA5	1.22×

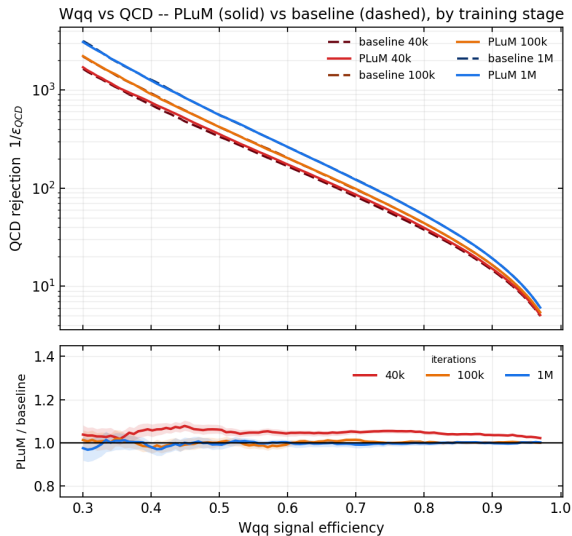
Features computed live by flashjet inside the loop, on the batch already resident on the GPU.

Parameters: 2.144 M baseline $\rightarrow$ 2.194 M with the PLuM feature block (+0.051 M).

Measured before the GPU-side collation change; the clustering itself is a small fraction of a step, so these are upper bounds on the true feature cost.

The cost is the reason a null result still matters: if a feature is free, a null is merely uninteresting — at **1.59×** it is a reason not to use it.

Backup — $W \rightarrow qq$, the fewest-splitting channel



$W \rightarrow qq$ has the **fewest** Lund splittings of any signal class (21.6/jet), yet shows the same early gain.

The effect is **not** confined to complex or b -containing jets — which is what a decay-structure explanation would predict.

Backup — two kinds of validation

First: toys, where truth is known

On generated inputs the *answer* is known analytically — so any observable landing off the predicted curve is a real bug, with nothing to hide behind.

Input A — toy fat jets — QCD-like and W-like fat jets

Input B — toy parton shower — leading-log parton shower

Then: real data, against FastJet

On detector-level jets there is no analytic answer — so the reference is **FastJet on the same events**, and the experiment's own reconstructed jets.

Input C — ATLAS Open Data — large- R jets and full events

The order matters: **toys first** pins the algorithm against a known answer; **real data second** shows it survives detector-level input. A failure at either stage means something different.

Backup — the validation ladder

rung	input	reference	result
unit tests	—	independent NumPy tree-walks	129 pass, incl. GPU paths
anti- k_t jet areas	toy B	the anti- k_t paper's own construction	rigid cones, $\rightarrow \pi R^2$
k_t substructure	toy A	known 2-prong vs QCD truth	clean separation
soft-drop z_g	toy B	leading-log $1/z$ prediction	on the curve
groomed mass vs β	toy B	Soft Drop Figs. 3–4	ordering reproduced
Lund plane	toy A	primary-Lund construction	hard split where predicted
jet-by-jet kinematics	real C	FastJet, same events	150/150 points, 100 % n_{jets}
full-event clustering	real C	the experiment's own jets	100 % match, ~ 600 clusters/ev

Each rung is closed against something independent — an analytic prediction, a separate NumPy implementation, or FastJet itself. No rung is validated against another part of flashjet.

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- L. Gouskos, B. Maier, *Particle-Lund Multimodality in Jet Taggers*, arXiv:**2605.26821**.
- H. Qu, L. Gouskos, *Jet tagging via particle clouds* (ParticleNet), Phys. Rev. D **101** (2020) 056019, arXiv:1902.08570.
- F. A. Dreyer, G. P. Salam, G. Soyez, *The Lund Jet Plane*, JHEP **12** (2018) 064, arXiv:1807.04758.
- F. A. Dreyer, H. Qu, *Jet tagging in the Lund plane with graph networks*, JHEP **03** (2021) 052, arXiv:2012.08526.
- A. J. Larkoski, S. Marzani, G. Soyez, J. Thaler, *Soft Drop*, JHEP **05** (2014) 146, arXiv:1402.2657.
- M. Cacciari, G. P. Salam, G. Soyez, *The anti- k_t jet clustering algorithm*, JHEP **04** (2008) 063, arXiv:0802.1189; *FastJet user manual*, Eur. Phys. J. C **72** (2012) 1896, arXiv:1111.6097.
- S. Catani, Yu. L. Dokshitzer, M. H. Seymour, B. R. Webber, Nucl. Phys. B **406** (1993) 187 (k_t); Yu. L. Dokshitzer *et al.*, hep-ph/9707323 and M. Wobisch, T. Wengler, hep-ph/9907280 (Cambridge/Aachen).
- J. de Favereau *et al.*, *DELPHES 3*, JHEP **02** (2014) 057, arXiv:1307.6346.

All benchmark data is **ATLAS Open Data** (CC0), read over the public redirector — no grid certificate required.

dataset	used for
ATLAS Top Tagging Open Data	timing + FastJet agreement
2020 ATLAS Jet Reconstruction	closure vs the experiment's own jets

Constituents are calorimeter clusters, massless. Physics closures use toy showers compared against leading-log analytic predictions, so the prediction being tested is unambiguous.